

Q1. Smart Hospital Patient Registration System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

public class Main {
    static ArrayList<String> patientIds = new ArrayList<>();
    static HashMap<String, String[]> patientMap = new HashMap<>();
    static HashMap<String, Integer> deptCount = new HashMap<>();

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice = -1;

        while (choice != 5) {
            if (!sc.hasNextLine()) break;
            choice = Integer.parseInt(sc.nextLine().trim());

            switch (choice) {
                case 1:
                    String id = sc.nextLine().trim();

                    if (patientMap.containsKey(id)) {
                        System.out.println("Duplicate Patient ID. Registration Failed.");
                        break;
                    }

                    if (patientIds.size() >= 100) {
                        System.out.println("Maximum Patient limit Reached.");
                        break;
                    }

                    String name = sc.nextLine().trim();

                    if (!name.matches("[a-zA-Z ]+")) {
                        System.out.println("Invalid Name. Only alphabets allowed.");
                        break;
                    }

                    String mobile = sc.nextLine().trim();

                    if (!mobile.matches("\\d{10}")) {
                        System.out.println("Invalid Mobile Number. Must be 10 digits.");
                        break;
                    }

                    String dept = sc.nextLine().trim().toUpperCase();

                    String doctor = sc.nextLine().trim();

                    String properName = toProperCase(name);

                    String[] record = {id, properName, mobile, dept, doctor};
                    patientIds.add(id);
                    patientMap.put(id, record);

                    deptCount.put(dept, deptCount.getOrDefault(dept, 0) + 1);

                    System.out.println("Patient Registered Successfully.");
                    break;

                case 2:
                    String searchId = sc.nextLine().trim();
```

```

        if (patientMap.containsKey(searchId)) {
            String[] p = patientMap.get(searchId);
            System.out.println("Patient Details");
            System.out.println("ID : " + p[0]);
            System.out.println("Name : " + p[1]);
            System.out.println("Mobile : " + p[2]);
            System.out.println("Department : " + p[3]);
            System.out.println("Doctor : " + p[4]);
        } else {
            System.out.println("Patient Not Found");
        }
        break;

    case 3:
        System.out.println("-----");
        System.out.println("ID\tName\tMobile\tDepartment\tDoctor");
        System.out.println("-----");
        for (String pid : patientIds) {
            String[] p = patientMap.get(pid);
            System.out.println(p[0] + " " + p[1] + " " + p[2] + " " + p[3] + " " + p[4]);
        }
        break;

    case 4:
        System.out.println("Department Count");
        for (Map.Entry<String, Integer> entry : deptCount.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
        break;

    case 5:
        break;

    default:
        System.out.println("Invalid Choice.");
    }
}
sc.close();
}

public static String toProperCase(String input) {
    String[] words = input.trim().split("\\s+");
    StringBuilder result = new StringBuilder();
    for (String w : words) {
        if (w.length() > 0) {
            result.append(Character.toUpperCase(w.charAt(0)))
                .append(w.substring(1).toLowerCase())
                .append(" ");
        }
    }
    return result.toString().trim();
}
}
}

```

INPUT

```

1
p101
Rahul
9121381941
Cardiology
Dr. Arun
3
4
2
P101
5

```

Q2. Online Library Book Management System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

public class Main {

    static class Book {
        String bookId;
        String title;
        String author;
        String category;
        String publisher;

        Book(String bookId, String title, String author, String category, String publisher) {
            this.bookId = bookId;
            this.title = title;
            this.author = author;
            this.category = category;
            this.publisher = publisher;
        }
    }

    static String toTitleCase(String s) {
        s = s.trim().toLowerCase();
        if (s.isEmpty()) return s;

        StringBuilder sb = new StringBuilder();
        String[] parts = s.split("\\s+");
        for (int i = 0; i < parts.length; i++) {
            String w = parts[i];
            if (w.length() == 0) continue;
            sb.append(Character.toUpperCase(w.charAt(0))).append(w.substring(1));
            if (i < parts.length - 1) sb.append(" ");
        }
        return sb.toString();
    }

    static String readNonEmpty(Scanner sc) {
        String s = sc.nextLine().trim();
        while (s.isEmpty()) {
            s = sc.nextLine().trim();
        }
        return s;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<Book> books = new ArrayList<>(500);
        HashMap<String, Book> bookById = new HashMap<>();
        HashSet<String> uniqueAuthors = new HashSet<>();
        HashMap<String, Integer> categoryCount = new HashMap<>();

        while (true) {
            System.out.println("----- Online library Book Management System -----");
            System.out.println("1. Add Book");
            System.out.println("2. Search Book by ID");
            System.out.println("3. Display All Books");
            System.out.println("4. Display Unique Authors");
            System.out.println("5. Category-wise Book Count");
            System.out.println("6. Exit");
            System.out.print("Enter choice: ");
```

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CODE

```
import java.util.*;

public class Main {

    static class Book {
        String bookId;
        String title;
        String author;
        String category;
        String publisher;

        Book(String bookId, String title, String author, String category, String publisher) {
            this.bookId = bookId;
            this.title = title;
            this.author = author;
            this.category = category;
            this.publisher = publisher;
        }
    }

    static String toTitleCase(String s) {
        s = s.trim().toLowerCase();
        if (s.isEmpty()) return s;

        StringBuilder sb = new StringBuilder();
        String[] parts = s.split("\\s+");
        for (int i = 0; i < parts.length; i++) {
            String w = parts[i];
            if (w.length() == 0) continue;
            sb.append(Character.toUpperCase(w.charAt(0))).append(w.substring(1));
            if (i < parts.length - 1) sb.append(" ");
        }
        return sb.toString();
    }

    static String readNonEmpty(Scanner sc) {
        String s = sc.nextLine().trim();
        while (s.isEmpty()) {
            s = sc.nextLine().trim();
        }
        return s;
    }

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